

HyDE Recommendation Engine — Production Plan (MVP)

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Status: Draft (Production Plan)

Last updated: 2026-01-26

Scope: HyDE Core Recommendation Engine + Customizations for Feed + Course Recommendation

Core rule (non-negotiable):

HyDE / LLM is allowed **only** in offline or refresh pipelines.

Online serving is fully deterministic and MUST NOT call LLM.

1) Executive Summary

This page defines the **production plan** (work breakdown, timeline, and ownership) to deliver:

- **HyDE Core Recommendation Engine** (shared engine)
 - 2 sprints (40 MDs, 2 DSs)
- **Feed Recommendation customization**
(feed-specific pipeline, subscores, aggregation)
 - 1.5 sprints (15 MDs, 1 DS)
- **Course Recommendation customization**
(course-specific pipeline, subscores, post-processing)
 - 1.5 sprints (15 MDs, 1 DS)

The plan is structured into:

- High-level plan (phases and major deliverables)
- Detailed plan (Jira-ready tasks)
- RACI ownership
- Milestones & acceptance criteria
- Operational & release readiness checklist
- Risks & mitigations

HyDE Course Recommendation approach – Details

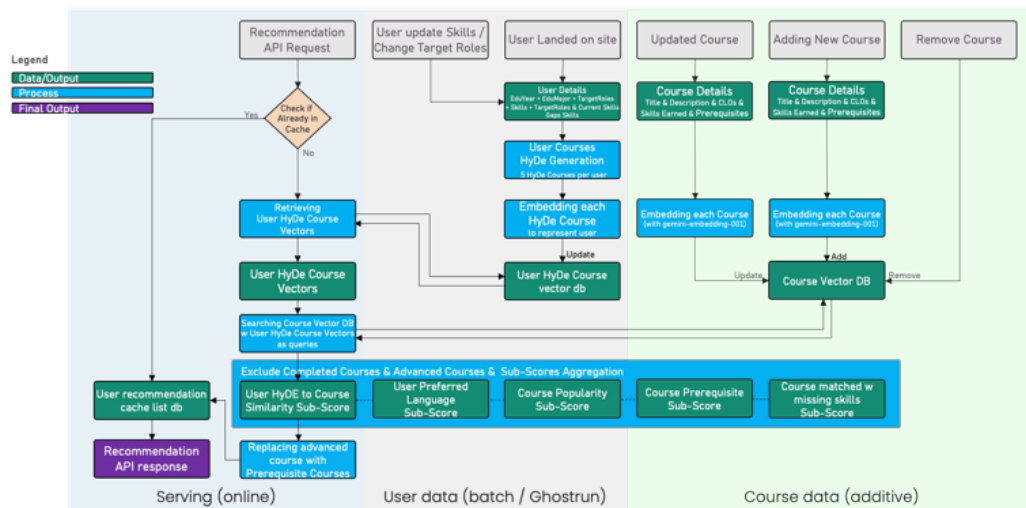


Figure1: Course Recommendation

HyDE Feed Recommendation approach & POC

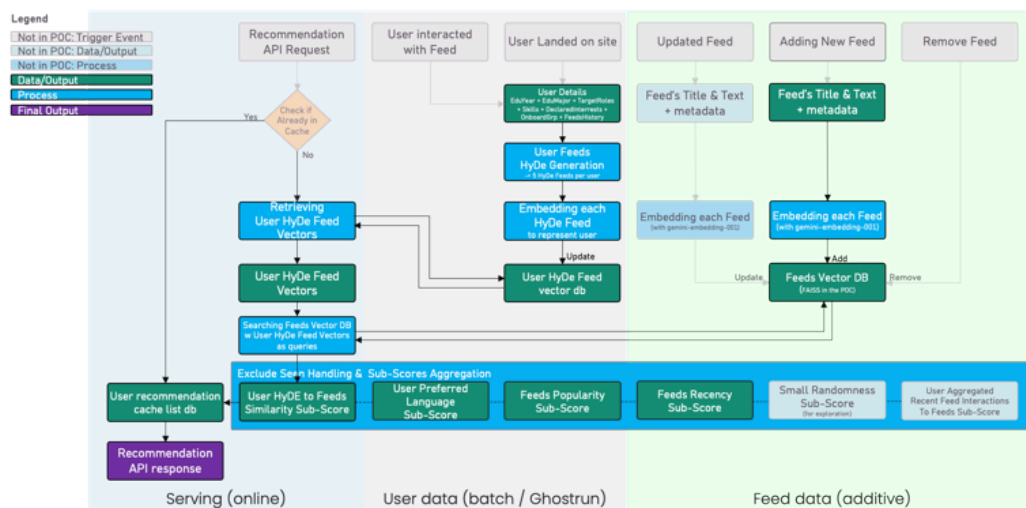


Figure2: Feed Recommendation

2) Guiding Principles (Production Guardrails)

2.1 LLM Boundary (Non-Negotiable)

LLM allowed only in:

- Batch HyDE generation (`pipeline_1_user_hyde`)
- User refresh (`pipeline_2_user_hyde_refresh`) — refresh only

LLM forbidden in:

- Online retrieval & serving (`pipeline_3_online_retrieval` and serving endpoints)
- Any module imported by the serving path

CI must fail if LLM clients are imported anywhere in the serving dependency tree.

2.2 Determinism Guarantees

- Same inputs ⇒ **stable outputs** (explicit tie-breakers enforced)
 - Cached HyDE embeddings (`.npy`) are **required** for serving
→ missing cache = **HARD FAIL**
 - Score aggregation weights are **config-driven (YAML)** — no hidden logic
-

2.3 Production Portability

- POC uses **FAISS**
 - Production target: **Vertex AI Vector Search**
 - Serving logic (`pipeline_3_online_retrieval`) must remain unchanged
→ only the **index adapter** is replaceable
-

3) Scope & Deliverables

3.1 HyDE Core Recommendation Engine (Shared)

Deliverables

- Item pipeline (store / update / remove / retrieve)
 - User HyDE pipeline (generate / embed / store / retire / retrieve)
 - Online serving pipeline (retrieve candidates → exclude-seen → score → rank)
 - HyDE orchestration pipeline (thin orchestration + structured logging)
 - Testing & validation (unit / integration / performance + gold metrics)
-

3.2 Feed Recommendation Customization

Deliverables

- Feed behavior gathering pipeline + feed vector DB deployment

- Feed-specific subscores and aggregation weights
 - End-to-end feed recommendation endpoints + documentation
 - Performance metrics (latency, MRR@k, P@k, Hit@k)
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3.3 Course Recommendation Customization

(Without skill gap analysis)

Deliverables

- Course behavior gathering pipeline + course vector DB deployment
 - Course-specific subscores + post-processing (prerequisite replacement)
 - End-to-end course recommendation endpoints + documentation
 - Performance metrics (latency, MRR@k, P@k, Hit@k)
-

4) Timeline Overview (High-Level)

Phase A — HyDE Core Engine Build

Duration: 2 sprints (40 MDs, 2 DSs)

Timeline: 2026-01-19 → 2026-02-13

- Items Pipeline (Jan 19–23)
 - User HyDE Pipeline (Jan 19–23)
 - Online Serving Pipeline (`pipeline_3_online_retrieval`) (Jan 26–30)
 - HyDE Orchestration (Jan 26–30)
 - Core Testing / Validation / Improvements (Feb 02–06)
 - Buffer (Feb 09–13)
-

Phase B — Customization for Feed & Course

Duration: 1.5 sprints each (15 MDs per track)

Timeline: 2026-02-13 → 2026-03-06

- Feed pipeline + scoring + integration + performance + docs
- Course pipeline + scoring + post-processing + integration + performance + docs
- UI + non-AI backend + SIT/UAT (separate track; dates TBD)

Feeds & Courses Recommendation: RACI (High LV)

Non-data related job

Task	Status	Estimated MCO	Start Date	Due Date	Terra DS	Terra DE	Terra Dev	Terra Business	S&L	HI	Stelligenc e	PIC
HyDE Core Recommendation Engine												
[Planning] Design overall solutions including tickets, timelines, and diagrams	Dev only total	40										
[Planning] Develop and deploy the recommendation engine	Done	5	6-Jan-26	16-Jan-26	R	I	I	C	-	A	-	Bloom
[Items Pipeline] Develop Shared Engine for store/update/remove items to be recommended	In progress	3.5	19-Jan-26	22-Jan-26	R/A	R	I	-	-	C	-	Shin
[Items Pipeline] Deploying, Testing, and documentation	In progress	1.5	22-Jan-26	23-Jan-26	R/A	R	I	-	-	C	-	Shin
[User HyOT Pipeline] Develop engine for creating user representations	In progress	3.5	19-Jan-26	22-Jan-26	R/A	R	I	-	-	C	-	Tun
[User HyOT Pipeline] Deploying, Testing, and documentation	In progress	1.5	22-Jan-26	23-Jan-26	R/A	R	I	-	-	C	-	Tun
[Online Serving Pipeline] Develop engine for online serving recommendation list	Not started	2.5	26-Jan-26	28-Jan-26	R/A	R	I	-	-	C	-	Tun
[Online Serving Pipeline] Deploying, Testing, and documentation	Not started	1.5	29-Jan-26	29-Jan-26	R/A	R	I	-	-	C	-	Tun
[HyDE Orchestration Pipeline] Develop Orchestrator Engine	Not started	4	26-Jan-26	29-Jan-26	R/A	R	I	-	-	C	-	Shin
[HyDE Orchestration Pipeline] Deploying, Testing, and documentation	Not started	2	30-Jan-26	30-Jan-26	R/A	R	I	-	-	C	-	Shin/Tun
[HyDE Core] Testing, Validation & Improvement	Not started	10	2-Feb-26	6-Feb-26	R/A	R	I	-	-	C	-	Shin/Tun
[HyDE Core] Buffer	Not started	10	9-Feb-26	13-Feb-26	R/A	R	I	-	-	C	-	Shin/Tun
Customize for Feed Recommendation												
[Feed Recommendation] Design solution for Feed Recommendation	Done	1	14-Jan-26	14-Jan-26	R	I	I	C	-	A	-	Bloom
[Feed Recommendation] Develop Custom Pipeline for Feed Recommendation	Not started	6	16-Feb-26	27-Feb-26	R/A	R	C	-	-	C	-	Tun
[Feed Recommendation] Develop the recent feeds sub-score, to handle recommendation adjustment while waiting for hyde update (to reduce LLM usage) *** Can be excluded if user representation don't need to be updated when waiting for hyde update i.e. Daily if user behavior update is ok	Not started (Optional)	5	NA	NA	R/A	C	C	C	-	C	-	NA
[Feed Recommendation] Deploying, Testing, and documentation	Not started	3.5	02-Mar-26	06-Mar-26	R/A	R	I	-	-	C	-	Tun
[Feed Recommendation] UI and non-AI backend development	Non-data	NA	NA	NA	C	C	R/A	C	-	C	-	NA
[Feed Recommendation] Integration Test (SIT/UAT)	Not started	NA	NA	NA	R	R	R/A	I	-	C	-	NA
Customize for Course Recommendation (without skill gap analysis)												
[Course Recommendation] Design solution for Course Recommendation	Done	1	14-Jan-26	14-Jan-26	R	I	I	C	-	A	-	Bloom
[Course Recommendation] Develop Custom Pipeline for Course Recommendation	Not started	6.5	16-Feb-26	27-Feb-26	R/A	R	R	-	I	C	-	Shin
[Course Recommendation] Deploying, Testing, and documentation	Not started	3.5	02-Mar-26	06-Mar-26	R	R	I	-	-	C	-	Shin
[Course Recommendation] UI and non-AI backend development	Non-data	NA	NA	NA	C	I	I	C	R/A	C	-	NA
[Feed Recommendation] Integration Test (SIT/UAT)	Not started	NA	NA	NA	R	R	R/A	I	-	C	-	NA

Figure 3: High LV timeline

5) Jira-Ready High-Level Ticket Plan

Feeds & Courses Recommendation: RACI (Details)

Task	Status	Estimated MCO	Start Date	Due Date	Terra DS	Terra DE	Terra Dev	Terra Business	S&L	HI	Stelligenc e	PIC
HyDE Core Recommendation Engine												
[Planning] Design overall solutions including tickets, timelines, and diagrams	Dev only total	40										
[Planning] Develop and deploy the recommendation engine	Done	5	6-Jan-26	16-Jan-26	R	I	I	C	-	A	-	Bloom
[Items Pipeline] Develop Shared Engine for store/update/remove items to be recommended	In progress	3.5	19-Jan-26	22-Jan-26	R/A	R	I	-	-	C	-	Shin
[Items Pipeline] Deploying, Testing, and documentation	In progress	1.5	22-Jan-26	23-Jan-26	R/A	R	I	-	-	C	-	Shin
[User HyOT Pipeline] Develop engine for creating user representations	In progress	3.5	19-Jan-26	22-Jan-26	R/A	R	I	-	-	C	-	Tun
[User HyOT Pipeline] Deploying, Testing, and documentation	In progress	1.5	22-Jan-26	23-Jan-26	R/A	R	I	-	-	C	-	Tun
[Online Serving Pipeline] Develop engine for online serving recommendation list	Not started	2.5	26-Jan-26	28-Jan-26	R/A	R	I	-	-	C	-	Tun
[Online Serving Pipeline] Deploying, Testing, and documentation	Not started	1.5	29-Jan-26	29-Jan-26	R/A	R	I	-	-	C	-	Tun
[HyDE Orchestration Pipeline] Develop Orchestrator Engine	Not started	4	26-Jan-26	29-Jan-26	R/A	R	I	-	-	C	-	Shin
[HyDE Orchestration Pipeline] Deploying, Testing, and documentation	Not started	2	30-Jan-26	30-Jan-26	R/A	R	I	-	-	C	-	Shin/Tun
[HyDE Core] Testing, Validation & Improvement	Not started	10	2-Feb-26	6-Feb-26	R/A	R	I	-	-	C	-	Shin/Tun
[HyDE Core] Buffer	Not started	10	9-Feb-26	13-Feb-26	R/A	R	I	-	-	C	-	Shin/Tun
Customize for Feed Recommendation												
[Feed Recommendation] Design solution for Feed Recommendation	Done	1	14-Jan-26	14-Jan-26	R	I	I	C	-	A	-	Bloom
[Feed Recommendation] Develop Custom Pipeline for Feed Recommendation	Not started	6	16-Feb-26	27-Feb-26	R/A	R	C	-	-	C	-	Tun
[Feed Recommendation] Develop the recent feeds sub-score, to handle recommendation adjustment while waiting for hyde update (to reduce LLM usage) *** Can be excluded if user representation don't need to be updated when waiting for hyde update i.e. Daily if user behavior update is ok	Not started (Optional)	5	NA	NA	R/A	C	C	C	-	C	-	NA
[Feed Recommendation] Deploying, Testing, and documentation	Not started	3.5	02-Mar-26	06-Mar-26	R/A	R	I	-	-	C	-	Tun
[Feed Recommendation] UI and non-AI backend development	Non-data	NA	NA	NA	C	C	R/A	C	-	C	-	NA
[Feed Recommendation] Integration Test (SIT/UAT)	Not started	NA	NA	NA	R	R	R/A	I	-	C	-	NA
Customize for Course Recommendation (without skill gap analysis)												
[Course Recommendation] Design solution for Course Recommendation	Done	1	14-Jan-26	14-Jan-26	R	I	I	C	-	A	-	Bloom
[Course Recommendation] Develop Custom Pipeline for Course Recommendation	Not started	6.5	16-Feb-26	27-Feb-26	R/A	R	R	-	I	C	-	Shin
[Course Recommendation] Deploying, Testing, and documentation	Not started	3.5	02-Mar-26	06-Mar-26	R	R	I	-	-	C	-	Shin
[Course Recommendation] UI and non-AI backend development	Non-data	NA	NA	NA	C	I	I	C	R/A	C	-	NA
[Feed Recommendation] Integration Test (SIT/UAT)	Not started	NA	NA	NA	R	R	R/A	I	-	C	-	NA

Figure 4: Detail LV timeline

EPIC: HyDE Core Recommendation Engine (Shared)

STORY: Items Pipeline — Shared Item Storage & Retrieval Engine

Estimate: 3.5 MD

Timeline: 2026-01-19 → 2026-01-22

PIC: Shin

Description

Build a shared items pipeline responsible for storing, updating, removing, and retrieving recommendable items (feeds, courses, etc.).

This pipeline is the single source of truth for vector-searchable content used by all recommendation variants.

Acceptance Criteria

- Supports item upsert, update, delete, and retrieval
- Retrieval supports vector similarity (cosine / IP)
- Optional BM25 hybrid retrieval is feature-flagged, not mandatory for MVP
- Item metadata is stable and retrievable across runs

Notes

- Domain-agnostic (feed / course reuse)
- No LLM usage allowed

STORY: Items Pipeline — Deployment, Testing & Documentation

Estimate: 1.5 MD

Timeline: 2026-01-22 → 2026-01-23

PIC: Shin

Description

Deploy item endpoints, add unit tests, and document API contracts in Confluence.

Acceptance Criteria

- Endpoints deployed and reachable
- Unit tests cover both the happy path + failure cases
- Confluence includes request/response examples

STORY: User HyDE Pipeline — User Representation & HyDE Generation

Estimate: 3.5 MD

Timeline: 2026-01-19 → 2026-01-22

PIC: Tun

Description

Implement deterministic user context building, HyDE query generation via LLM, embedding, and caching as artifacts.

Acceptance Criteria

- User context generation is deterministic

- HyDE prompts are YAML-controlled
- Query embeddings cached as `.npy` and referenced in bundle JSON

Notes

- Signature comparison required for skip/refresh
 - Strict language enforcement (TH / EN)
-

STORY: User HyDE Pipeline — Deployment, Testing & Documentation

Estimate: 1.5 MD

Timeline: 2026-01-22 → 2026-01-23

PIC: Tun

Description

Deploy user HyDE endpoints and validate refresh logic.

Acceptance Criteria

- Refresh skips when the signature is unchanged
 - Tests confirm LLM is not called unnecessarily
 - Refresh policy documented in Confluence
-

STORY: Online Serving Pipeline — Deterministic Recommendation Serving

Estimate: 2.5 MD

Timeline: 2026-01-26 → 2026-01-28

PIC: Tun

Description

Implement deterministic online serving (`pipeline_3_online_retrieval`) using cached user embeddings and vector retrieval.

Acceptance Criteria

- Uses cached `.npy` embeddings only
- Exclude-seen logic applies a N-day (i.e. 30 days) window
- Output includes subscores, final score, and debug fields
- Response schema is versioned and API-stable

Notes

- Missing cached embeddings = HARD FAIL
 - Deterministic tie-break rules enforced
-

STORY: Online Serving Pipeline — Deployment, Testing & Documentation

Estimate: 1.5 MD

Timeline: 2026-01-28 → 2026-01-30

PIC: Tun

Description

Deploy serving endpoints and enforce “no-LLM-in-serving” via CI.

Acceptance Criteria

- End-to-end integration tests pass
 - CI blocks LLM imports in the serving path
 - Serving API documented in Confluence
-

STORY: HyDE Orchestration Pipeline — Orchestrator Engine

Estimate: 4 MD

Timeline: 2026-01-26 → 2026-01-29

PIC: Shin

Description

Build a thin orchestration layer to trigger item refresh, user refresh, and diagnostics.

Acceptance Criteria

- Safe refresh triggering
 - Structured logs include request_id and timings
 - Failures are observable and actionable
-

STORY: HyDE Orchestration — Deployment, Testing & Documentation

Estimate: 2 MD

Timeline: 2026-01-30

PIC: Shin

Description

Deploy the orchestrator and validate full-pipeline execution.

Acceptance Criteria

- Full pipeline test passes via orchestrator
 - Logs show step-by-step execution
 - Confluence documentation complete
-

EPIC: HyDE Core — Testing, Validation & Improvement

Estimate: 10 MD

Timeline: 2026-02-02 → 2026-02-06

PIC: Shin / Tun

Description

Stabilize the HyDE core through regression testing, gold validation, performance tuning, and fixes.

Acceptance Criteria

- All unit and integration tests pass
 - Gold metrics (MRR@k, P@k, Hit@k) recorded
 - Determinism and latency targets met
-

EPIC: HyDE Core — Buffer

Estimate: 10 MD

Timeline: 2026-02-09 → 2026-02-13

PIC: Shin / Tun

Description

Buffer to ensure the core engine is ready before proceeding

6) Jira-Ready Detailed Ticket Plan — Customization Layers

6A) Feed Recommendation Customization

STORY: Feed Behavior Data Gathering Pipeline

Estimate: 3 MD

Timeline: 2026-02-16 → 2026-02-18

PIC: Tun

Description

Collect and normalize on-site user behavior related to feed consumption.

Acceptance Criteria

- Events normalized and time-windowed
 - Schema aligns with exclude-seen logic
 - Data is reproducible and testable
-

STORY: Feed Data Synthesis & Feed Vector DB Deployment

Estimate: 2 MD

Timeline: 2026-02-19 → 2026-02-20

PIC: Tun

Description

Synthesize feed content, attach tags as metadata, and deploy feed vector index.

Acceptance Criteria

- Feed tags/categories are queryable as metadata
 - Index build reproducible
 - Compatible with the core retrieval engine
-

STORY: Feed Sub-Scores & Aggregation (Feed-Specific)

Estimate: 5 MD

Timeline: 2026-02-23 → 2026-02-27

PIC: Tun

Description

Implement feed-specific subscores and aggregation weights.

Acceptance Criteria

- Subscores deterministic
 - Weights YAML-driven
 - Optional “recent feed” logic is safely toggleable
-

STORY: Feed Recommendation — Integration, Testing & Performance

Estimate: 4 MD

Timeline: 2026-03-02 → 2026-03-05

PIC: Tun

Description

Integrate feed customization with the core engine and run full validation.

Acceptance Criteria

- End-to-end feed recommendation works
 - Latency and quality metrics recorded
 - Results are explainable and stable
-

STORY: Feed Recommendation — Documentation & API Spec

Estimate: 1 MD

Timeline: 2026-03-06

PIC: Tun

Description

Document feed recommendation APIs and configuration.

Acceptance Criteria

- API examples included
 - Scoring logic explained
 - Ops notes provided
-

6B) Course Recommendation Customization - *(No Skill Gap Analysis)*

STORY: Course Behavior Data Gathering Pipeline

Estimate: 3 MD

Timeline: 2026-02-16 → 2026-02-18

PIC: Shin

Description

Collect and normalize user behavior related to course interactions.

Acceptance Criteria

- Events normalized

- Compatible with exclude-seen logic
 - Deterministic aggregation
-

STORY: Course Vector DB Deployment

Estimate: 2 MD

Timeline: 2026-02-19 → 2026-02-20

PIC: Shin

Description

Deploy course content into a vector database.

Acceptance Criteria

- Course embeddings indexed successfully
 - Metadata retrievable
 - Serving compatibility verified
-

STORY: Course Sub-Scores, Aggregation & Post-Processing

Estimate: 5 MD

Timeline: 2026-02-23 → 2026-02-27

PIC: Shin

Description

Implement course-specific subscores and prerequisite replacement logic.

Acceptance Criteria

- Alignment logic deterministic
 - Prerequisite replacement validated
 - YAML-configurable weights
-

STORY: Course Recommendation — Integration, Testing & Performance

Estimate: 4 MD

Timeline: 2026-03-02 → 2026-03-05

PIC: Shin

Description

Integrate the full course recommendation pipeline and validate.

Acceptance Criteria

- End-to-end pipeline passes
 - Metrics recorded
 - Compatible with the core retrieval engine
-

STORY: Course Recommendation — Documentation & API Spec

Estimate: 1 MD

Timeline: 2026-03-06

PIC: Shin

Description

Document course recommendation APIs and logic.

Acceptance Criteria

- API contracts documented
 - Post-processing clearly explained
 - Ready for SIT/UAT handoff
-

7) High-Level RACI Matrix

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

- **Terra DS:** R / A (core pipelines)
- **Terra DE:** R (data & vector store)
- **Terra Dev:** I / C
- **Terra Business:** C
- **SKL:** A / C
- **HI:** C
- **STelligence:** –

PIC Mapping

- Bloom: planning & solution definition
 - Shin: item pipeline, orchestration, serving
 - Tun: user HyDE, feed & course customization
-

8) Milestones & Release Criteria

M1 — Core Engine Ready (2026-02-13)

- Items, User HyDE, Serving, Orchestration deployed
- Gold validation + determinism tests pass
- “No LLM in serving” enforced

M2 — Feed Recommendation Ready (2026-03-06)

- Feed vector DB built
- Subscores & aggregation tuned
- Performance baseline recorded
- Docs completed

M3 — Course Recommendation Ready (2026-03-06)

- Course vector DB built
- Prerequisite logic validated
- Performance baseline recorded
- Docs completed

9) Operational Readiness Checklist

9.1 Deployment

- Environments defined (dev / staging / prod)
- Configs externalized (YAML + env)
- Rollback plan documented
- Logs / metrics / traces available

9.2 Safety & Compliance

- PII whitelist enforced
- LLM boundary tests in CI
- Refresh rate limits defined

9.3 Reliability

- Cold start behavior documented (Cloud Run / FAISS)
- Vector index load strategy defined (cache vs memory)
- Error budget & alert thresholds defined (p90 latency, error rate)

10) Risks & Mitigations

Risk	Impact	Mitigation
LLM called in serving path	High	CI import guard + mock LLM tests
Missing <code>.npy</code> user bundle (data-lake path)	High	Hard-fail + orchestration refresh or switch to vector DB
Cold start & index load latency (FAISS / Cloud Run)	Medium–High	Pre-warm, min instances, load-lock, async/lazy load
Vector backend swap (FAISS → Vertex)	Medium	Adapter-only change + golden tests

11) References

- **POC page:** *HyDE Feed Recommendation POC — Progress + Verified Run Output (2026-01-18)*
- **POC Repo:** [GitHub - nachai-l/poc_hyde_feed_recommendation: Deterministic HyDE-based feed recommendation POC with offline LLM query synthesis and fully LLM-free online retrieval and ranking. Designed for debuggability, explainability, and production-ready evaluation.](#)
- **Verified principles:** deterministic serving, cached HyDE embeddings, YAML scoring, gold validation + CI gates